

Week 4 Plan Explanation – Event Study (Earnings or Macro)

Week Goal

Design and execute a rigorous event study to test a market hypothesis.

By the end of the week, the intern can design an event study, compute AR, CAR, AAR, and CAAR correctly, test significance, and distinguish economic interpretation from statistical output.

Task 1: Event Study Methodology

This task defines the event–study design before any result is calculated, including the event date, estimation window, event window, and normal–return model.

Detailed Work

- What is an event study? History and applications
- Key concepts: event date, estimation window, event window
- Normal return models: market model, market–adjusted, mean–adjusted
- Abnormal return: $AR = R_{\text{actual}} - R_{\text{expected}}$
- Cumulative abnormal return (CAR)

Concrete Output

- Read 1–2 classic event study papers for methodology reference

Quality Check

The event and estimation windows must not overlap, the event–date convention must be documented, and the chosen normal–return model must match the hypothesis.

Task 2: Data Preparation

This task creates a valid event panel and ensures that each observation has enough pre–event history for estimation.

Detailed Work

- Choose event type:
 - Option A: Earnings announcements (earnings surprise)
 - Option B: FOMC meeting dates (rate decisions)
- Collect event dates (can use a provided CSV or scrape from public source)
- Define event window: $[-5, +5]$ trading days around the event
- Define estimation window: $[-260, -11]$ for market model estimation

Concrete Output

- Create event dataset: merge event dates with stock returns
- Filter: at least 200 days of estimation window data per event

Quality Check

Each event must pass the minimum–history filter, merges must use trading dates correctly, and duplicate or clustered events must be identified.

Task 3: Abnormal Return Computation

This task estimates expected returns and isolates event–window abnormal returns and cumulative abnormal returns.

Detailed Work

- Estimate market model: $R_{i,t} = \alpha + \beta * R_{m,t} + \varepsilon$ over estimation window
- Compute expected return in event window using estimated α and β
- Compute abnormal returns: $AR_{i,t} = R_{i,t} - (\hat{\alpha} + \hat{\beta} * R_{m,t})$

- Compute CAR over event window

Concrete Output

- Implement `estimate_market_model()` and `compute_abnormal_returns()` in notebook
- Spot-check a few events manually to verify calculations

Quality Check

Several events must be reconstructed manually, and abnormal returns must use coefficients estimated only from the pre-event window.

Task 4: Aggregation & Statistical Testing

This task aggregates evidence across events, quantifies uncertainty, and checks whether results differ across meaningful subsamples.

Detailed Work

- Average across events: AAR (average abnormal return) and CAAR (cumulative average abnormal return)
- Cross-sectional t-test: is AAR significantly different from zero?
- Plot CAAR from t-5 to t+5 with confidence bands
- Subsample analysis: by year, by sector, by surprise direction (positive vs negative)

Concrete Output

- Plot CAAR with 95% confidence interval
- Create a table of AAR and t-statistics for each day in the event window
- Split by positive vs negative surprise – do patterns differ?

Quality Check

Confidence intervals and t-statistics must use a stated method, while statistical significance must remain separate from economic magnitude.

Task 5: Interpretation & Write-Up

This task interprets the statistical output in economic terms while addressing alternative explanations and study limitations.

Detailed Work

- Interpret results: is there pre-event drift? Post-event drift? Reversal?
- Discuss limitations: event clustering, thin trading, confounding events
- Market efficiency implications

Concrete Output

- Write one-page summary of findings
- Friday demo: present CAAR plots and key conclusions

Quality Check

The write-up must address pre-event drift, post-event drift, confounding news, event clustering, and the limits of causal interpretation.

Week Deliverables

- `notebooks/week4/event_study.ipynb`
- One-page findings summary with CAAR plot and significance table
- Weekly memo

Recommended Reading

- MacKinlay (1997), "Event Studies in Economics and Finance"
- Ball & Brown (1968) – the original earnings event study
- Campbell, Lo, & MacKinlay, *The Econometrics of Financial Markets*, Chapter 4

Common Pitfalls

- Event date errors (using announcement date vs filing date)
- Not accounting for event clustering (events on the same date)
- Confusing statistical significance with economic significance
- Overlapping event and estimation windows

What Success Looks Like

The intern can design an event study, compute AR, CAR, AAR, and CAAR correctly, test significance, and distinguish economic interpretation from statistical output.